Experiment 21 :

Create a Python program that performs syntax-driven semantic analysis by extracting noun phrases and their meanings from a sentence.

Code:

```
import nltk
from nltk import word_tokenize, pos_tag, RegexpParser
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')
sentence = input("Enter a sentence: ")
words = word_tokenize(sentence)
tags = pos_tag(words)
grammar = "NP: {<DT>?<JJ>*<NN.*>+}"
chunk_parser = RegexpParser(grammar)
tree = chunk_parser.parse(tags)
print("Noun Phrases:")
for subtree in tree.subtrees():
    if subtree.label() == "NP":
        print(" ".join(word for word, tag in subtree.leaves()))
```

Output